

EXPERIMENT 22

Push and Pull Docker Images Using Docker Hub

1. Aim

To create a Docker image for a static HTML website, push it to Docker Hub, pull the image, and run it successfully as a Docker container.

2. Requirements

Docker, Docker Hub account, static HTML file, Dockerfile, and web browser.

3. Dockerfile

```
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

4. Build the Image

```
docker build -t docker-hub-demo:latest .
docker images
```

5. Tag and Push to Docker Hub

```
docker login
docker tag docker-hub-demo:latest <username>/docker-hub-demo:latest
docker push <username>/docker-hub-demo:latest
```

6. Pull and Run

```
docker pull <username>/docker-hub-demo:latest
docker run -d --name docker-hub-demo -p 8085:80 <username>/docker-hub-demo:latest
```

7. Testing

Open localhost:8085 in the browser. The static HTML website should be displayed successfully.

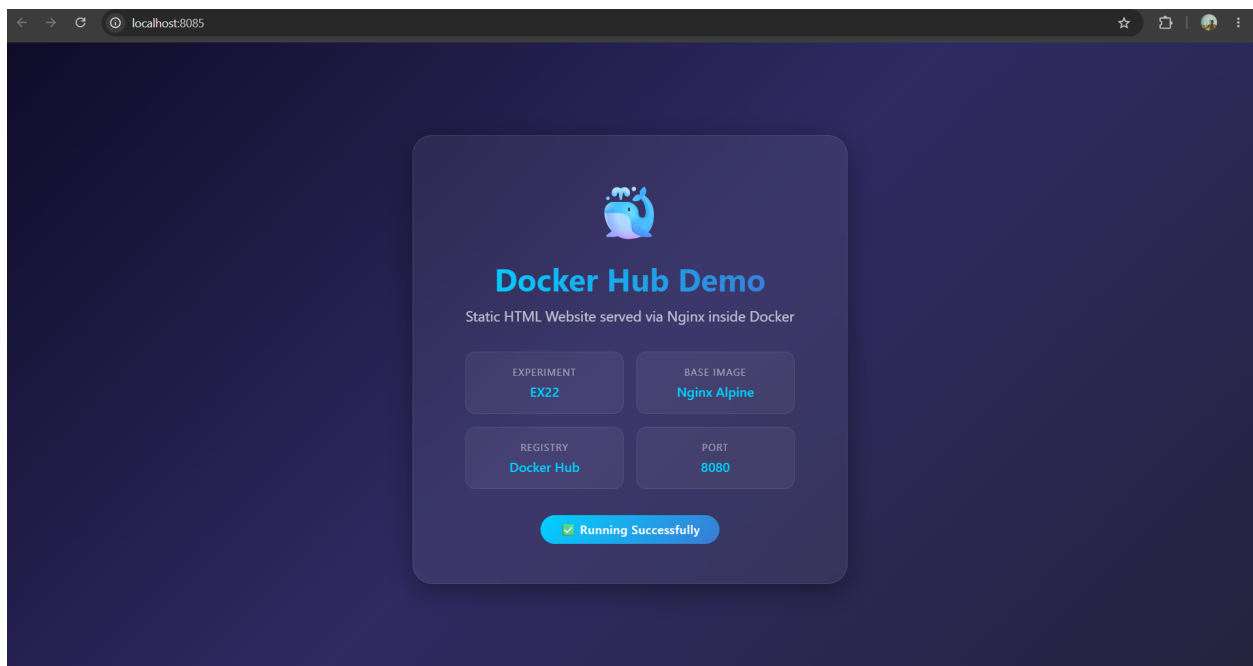


Figure 1: Docker Hub Demo successfully running at localhost:8085.

8. Result

The static HTML website was successfully built into a Docker image, published to Docker Hub, pulled, and run successfully. The browser test confirms that the containerized website is working.

9. Conclusion

Docker Hub provides a convenient way to store and distribute Docker images, allowing the same containerized application to be pulled and run on another machine.